Group members: Amy Chen, Yingying Chu, Mingyang Gao, Shiyu Wu

1. Abstract / Introduction

1.1 Dataset:
https://www.kaggle.com/datasets/sumanthvrao/daily-climate-time-series-data?resource=download

1.2 Goals and Objectives:
This project investigates the predictability of daily mean temperature in Delhi, India, using historical observations and three explanatory variables: humidity, wind speed, and mean pressure. The dataset spans from January 1, 2013 to April 24, 2017, and model performance is evaluated using out-of-sample RMSE based on a holdout test set.

We compare four classes of models: (1) simple baseline rules, (2) exponential smoothing, (3) ARIMA models without explanatory variables, and (4) ARIMAX models incorporating external predictors.

1.3 Main Conclusions:
Simple rules show that the average (IID) model performs poorly, while the persistence model performs much better, indicating strong short-term dependence. Exponential smoothing further improves accuracy, though it fails to capture the strong annual seasonality in the data.

Holt–Winters + ARIMA models without explanatory variables provide further improvement (RMSE ≈ 1.76) by modeling residual autocorrelation. In particular, the hybrid Holt–Winters + ARIMA(2,0,5) model achieves the best performance within this class.

ARIMAX models incorporating humidity and wind speed further reduce prediction error (RMSE ≈ 1.68), while mean pressure does not contribute significantly. The best performance is achieved by explicitly modeling seasonality using a sinusoidal term combined with ARIMA errors, yielding the lowest RMSE (≈ 1.64).

2. Exploratory Data Analysis (EDA)

2.1 Variable Description:

- ***meantemp***: Daily mean temperature computed from 3-hourly measurements (℃).
- ***humidity***: Daily humidity (g/m^3).
- ***wind_speed***: Daily wind speed (km/h).
- ***meanpressure***: Daily atmospheric pressure (atm).

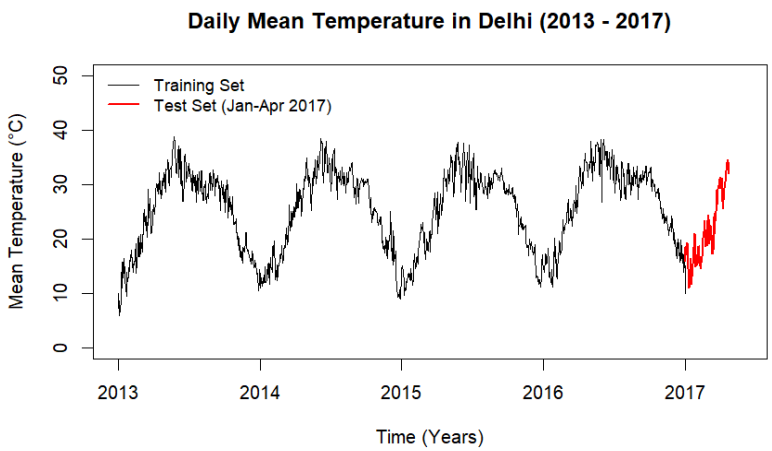
2.2 Summary Statistics:

```
--- Basic Summary Statistics ---
    meantemp        humidity        wind_speed      meanpressure
 Min.   : 6.00   Min.   : 13.43   Min.   : 0.000   Min.   :  -3.042
 1st Qu.:18.86   1st Qu.: 50.38   1st Qu.: 3.475   1st Qu.:1001.580
 Median :27.71   Median : 62.62   Median : 6.222   Median :1008.563
 Mean   :25.50   Mean   : 60.77   Mean   : 6.802   Mean   :1011.105
 3rd Qu.:31.31   3rd Qu.: 72.22   3rd Qu.: 9.238   3rd Qu.:1014.945
 Max.   :38.71   Max.   :100.00   Max.   :42.220   Max.   :7679.333

--- Standard Deviations ---
    meantemp     humidity   wind_speed meanpressure
    7.348103    16.769652     4.561602   180.231668
```

(Figure 2.21)

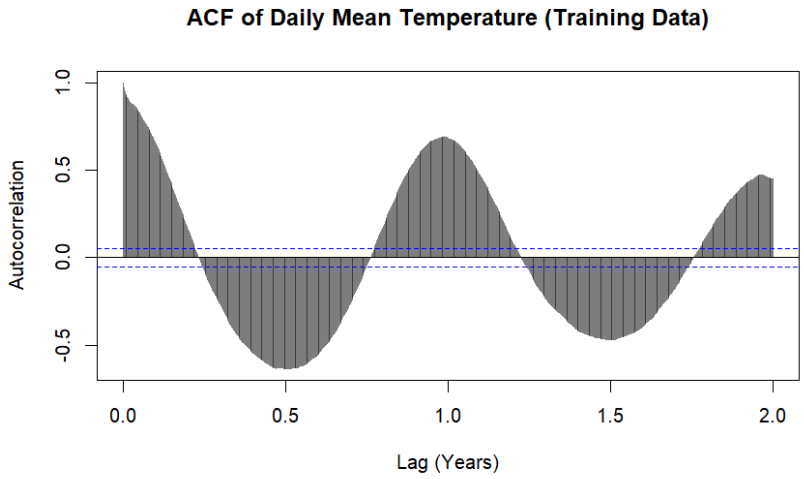
Basic summary statistics for all variables are shown in Figure 2.21, indicating that most variables fall within reasonable ranges. However, the *meanpressure* variable contains extreme outliers (Min: -3.0, Max: 7679.3). These were addressed during the data wrangling phase before fitting multivariate models.

2.3 Visual Diagnostics:



(Figure 2.31)

Figure 2.31 shows the time series plot of daily mean temperature from 2013 to 2017. Several important patterns can be observed. First, there is a clear and repeated seasonal pattern, with temperature rising and falling in a regular yearly cycle. Second, the variability appears relatively stable over time, with no obvious structural breaks.



(Figure 2.32)

Figure 2.32 presents the autocorrelation function (ACF) of the training data. Signs such as significant spikes at Lag 1 and Lag 2, as well as a strong oscillating pattern, confirm seasonality. The slow decay also suggests that the series is not stationary.

3. Time Series Forecasting Models

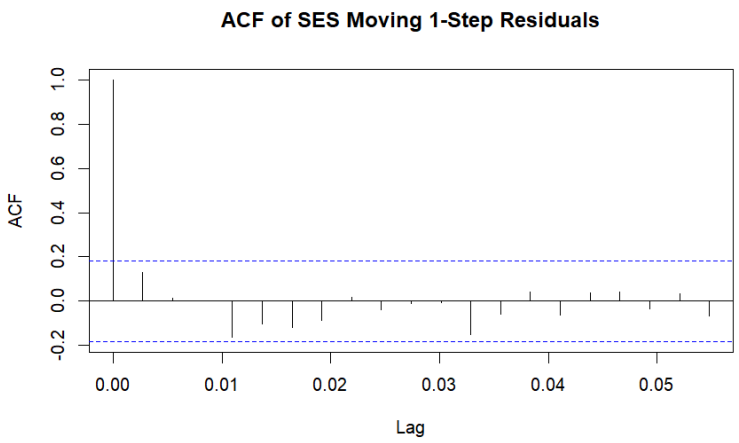
3.1 Baseline and Exponential Smoothing

We begin by evaluating simple baseline models and exponential smoothing methods using moving one-step-ahead forecasts on the test set.

	Model	RMSE
1	Mean (IID)	7.375808
2	Persistence	1.771308
3	SES (1-step moving)	1.754914
4	Holt-Winters (Additive)	1.930906

(Table 3.11)

The mean (IID) model, which predicts future values using the average of the training data, performs poorly because it ignores temporal dependence. In contrast, the persistence model, which uses the most recent observation as the forecast, performs substantially better. This indicates that daily temperature exhibits strong short-term dependence and that recent observations contain valuable predictive information. Exponential smoothing methods are then applied to capture different components of the time series and achieve the lowest RMSE of 1.75.



(Figure 3.12)

However, the ACF for SES residuals in Figure 3.12 indicates significant remaining autocorrelation at seasonal lags, meaning that, despite their numerical performance, smoothing methods fail to adequately capture the strong annual seasonal structure present in the data.

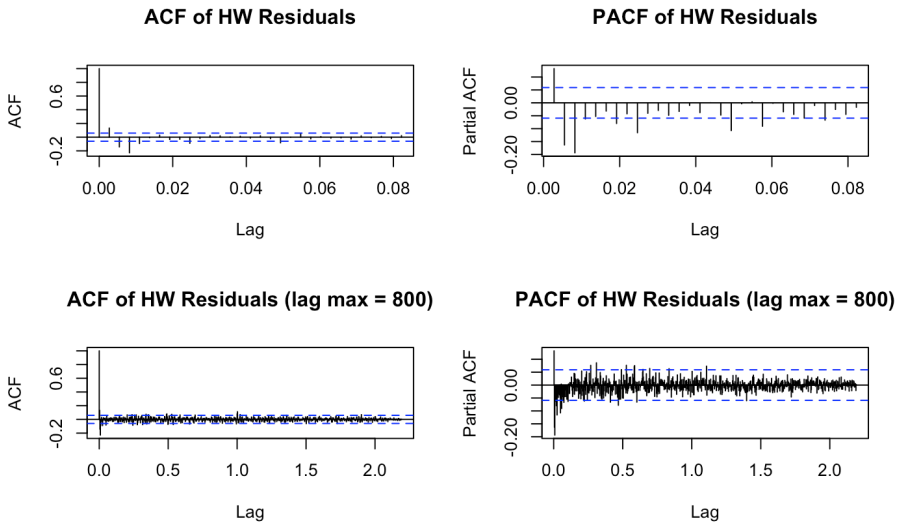
3.2 ARIMA Modeling

Because the daily temperature series exhibits a strong annual seasonal pattern (period = 365), applying a full seasonal ARIMA model is theoretically appropriate. Indeed, auto.arima on the original series suggested the model ARIMA(3,1,1)(0,1,0)[365].

However, due to the very large seasonal period, fitting seasonal ARIMA models directly is computationally challenging and unstable in practice. In particular, the implementation in R has a limitation on the maximum supported lag (maximum at lag 350), which prevents us from reliably estimating models with a seasonal period of 365.

To address this, we adopted a hybrid modeling approach:

1. Holt-Winters (additive) was first used to capture the dominant yearly seasonal pattern.
2. The resulting residual series was then modeled using non-seasonal ARIMA models to capture remaining short-term dependencies.



(Figure 3.21)

The ACF and PACF plots (Figure 3.21) of the Holt-Winters residuals are used to guide the ARIMA model selection. The ACF shows noticeable spikes up to around lag 4, after which most values fall within the 95% confidence interval, suggesting that a moving average component of approximately order 4 may be appropriate.

On the other hand, the PACF only shows significant spikes in the first one or two lags and then quickly becomes insignificant with no clear cutoff pattern, indicating that the autoregressive component is relatively weak and likely of low order.

Applying auto.arima to the Holt–Winters residuals suggested: ARIMA(0,0,5)

However, since models selected by AIC do not necessarily yield the best forecasting performance, we explored a range of nearby models by varying AR and MA orders. In addition, we also

considered models with first differencing (i.e., d=1) to account for any remaining non-stationarity in the residual series.

Candidate models included:

Each model was evaluated using moving 1-step ahead forecasts on the holdout set, and the corresponding holdout set RMSE values were computed. The results are summarized as follows:

Model	arima(0,0,1)	arima(0,0,2)	arima(0,0,3)	arima(0,0,4)	arima(0,0,5)	arima(0,0,6)	arima(1,0,0)
rmse	1.819130	1.827689	1.872307	1.843472	1.814377	1.789092	1.832949
Model	arima(2,0,0)	arima(1,0,1)	arima(1,0,5)	arima(3,0,3)	arima(3,1,3)	arima(2,0,5)	arima(2,1,5)
rmse	1.817366	1.816787	1.761159	1.765750	1.764433	1.760288	1.769221
Model	arima(3,0,5)	arima(3,0,4)					
rmse	1.768030	1.765558					

(Table 3.22)

The best-performing model is Holt–Winters + ARIMA(2,0,5) with an RMSE of 1.760.

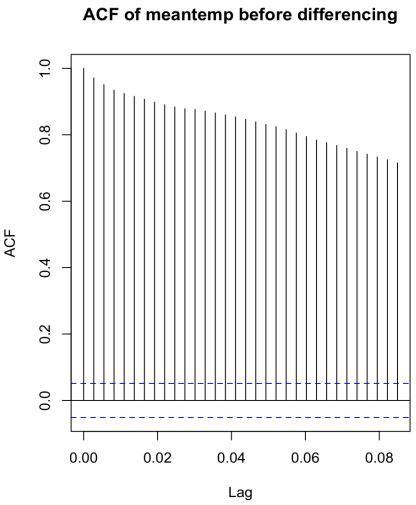
```
Call:
arima(x = hw_resid, order = c(2, 0, 5), include.mean = FALSE)

Coefficients:
         ar1      ar2      ma1      ma2      ma3     ma4     ma5
      0.9390  -0.0740  -0.8919  -0.1750  -0.0890  0.1119  0.0486
s.e.  0.3522   0.2896   0.3515   0.2762   0.0947  0.0546  0.0600

sigma^2 estimated as 3.046:  log likelihood = -2168.6,  aic = 4353.2

Training set error measures:
                     ME     RMSE      MAE      MPE     MAPE      MASE         ACF1
Training set 0.05155649 1.745166 1.222344 43.81992 310.6605 0.6512156 -0.001535485
```

(Table 3.23)

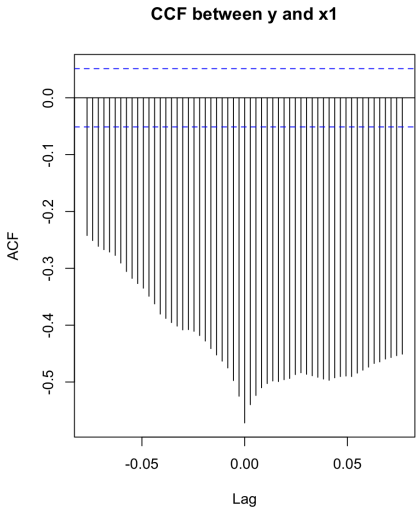
Notably, this result is consistent with the patterns observed in the ACF and PACF plots of the Holt–Winters residuals (Figure 3.21). The ACF shows a gradual cutoff around lag 4–5, suggesting a higher-order MA component, while the PACF does not display a clear cutoff, indicating a relatively small AR order. This aligns well with the selected ARIMA(2,0,5) model, which combines a low-order AR component with a higher-order MA component.

3.3 ARMAX Modeling

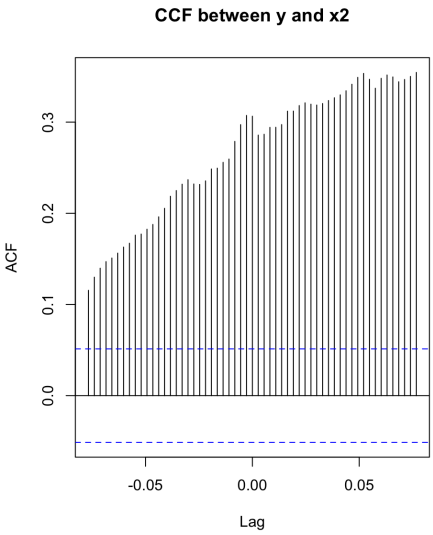
We consider the basic diagnostic plots, i.e., the ACF and PACF of the mean temperature (Y) before any treatment, as well as its CCF with humidity (X_1), wind speed (X_2), and pressure (X_3).



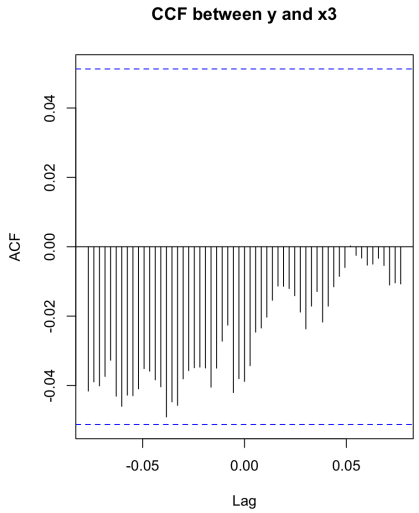
(Figure 3.31)



(Figure 3.32)



(Figure 3.33)



(Figure 3.34)

Figure 3.31 reveals that the mean temperature exhibits non-stationarity, indicated by the slowly decaying ACF. To address this, first differencing is applied to achieve stationarity. In addition, the original series shows a strong correlation with all three explanatory variables (Figures 3.32 to 3.34). Since the ARMAX framework models the relationship between the response variable and external predictors while accounting for autocorrelated errors, we consider a simplified ARMAX specification. Specifically, we use the previous day's values of X_1, X_2, and X_3 as linear predictors, combined with an ARIMA structure for the residuals.

To illustrate this approach, we first fit a linear regression model using the lagged predictors, and then model the resulting residuals with ARIMA. The summary of the fitted linear model is presented below:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 39.1873623  1.1976291  32.721  < 2e-16 ***
x1prev      -0.2214827  0.0103165 -21.469  < 2e-16 ***
x2prev       0.1569101  0.0379343   4.136 3.73e-05 ***
x3prev      -0.0012810  0.0008904  -1.439     0.15
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.133 on 1458 degrees of freedom
Multiple R-squared:  0.3035,    Adjusted R-squared:  0.3021
F-statistic: 211.8 on 3 and 1458 DF,  p-value: < 2.2e-16
```

(Figure 3.35)

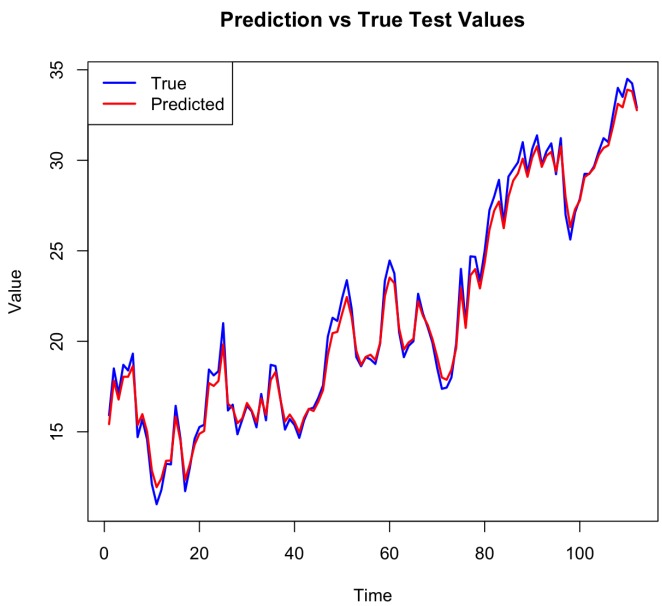
The initial regression results in Figure 3.35 indicate that X_3 is not a significant predictor. As a result, it is excluded from the model, and we proceed by regressing the response variable on X_1 and X_2 only. We then fit the residuals of the regression to an ARIMA process.

Auto ARIMA chooses an ARIMA(2,1,2) process. This is consistent with the preliminary analysis, which suggested the need for first differencing to achieve stationarity and showed significant ACF and PACF at the first few lags.

Based on these findings, we fit an ARIMAX model directly, specifying an ARIMA(2,1,2) residual and regressing on X_1 and X_2. The eventual test RMSE is 1.68. We provide a comparison of the fitted prediction and the true test value:

```
Series: resid_reg2
ARIMA(2,1,2)

Coefficients:
          ar1     ar2      ma1      ma2
      -0.3985  0.2754  -0.1345  -0.5920
s.e.   0.1628  0.0485   0.1641   0.1314

sigma^2 = 6.454:  log likelihood = -3433.52
AIC=6877.04   AICc=6877.08   BIC=6903.47
```

(Figure 3.36)



(Figure 3.37)

More sophisticated and engineered predictors:

1. Predict using linear combinations of the predictors, similar to interaction effects in regression models.

We introduced a mixed term involving X_1, X_2, and X_3, and then followed the suggested layout above. We called the linear model and the variable significance is as follows.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.995e+01  1.161e+01  -3.441 0.000595 ***
x1prev       1.278e+00  2.177e-01   5.870 5.38e-09 ***
x2prev      -8.016e-02  1.383e-01  -0.579 0.562353
x3prev       7.943e-02  1.144e-02   6.946 5.66e-12 ***
X1X2         3.747e-03  2.234e-03   1.677 0.093697 .
X1X3        -1.520e-03  2.147e-04  -7.080 2.24e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

(Figure 3.38)

Hence, we regress on X_1, X_3 and X_1X_3. We then fit an ARIMA on the residual. Auto ARIMA again selects ARIMA(2,1,2). This is not surprising since changing linear predictors would not affect the dependence result. The test RMSE is 1.689, which is slightly higher than the simpler model. This suggests that using mixed terms decreases models' ability to generalize.

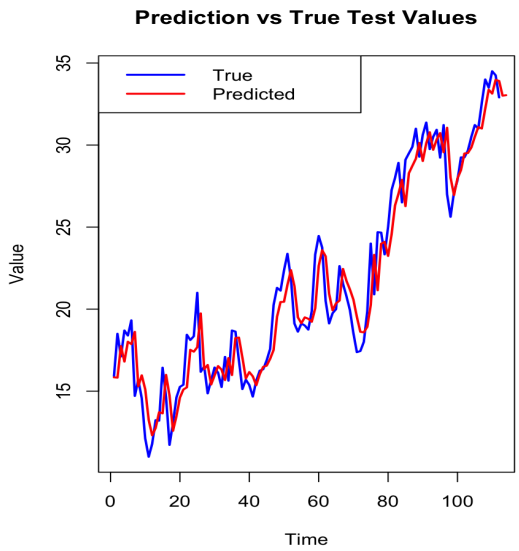
2. Incorporating Seasonality via a Sinusoidal Term.

Previous results indicate seasonality. However, our dataset records daily data, which makes fitting seasonal ARIMA difficult. Here, we use regression on a sinusoid with period = 365 to fit the seasonal pattern. The sinusoidal regression result is provided below.

```
lm(formula = y ~ cosine + sine, data = train_df)

Residuals:
     Min       1Q   Median       3Q      Max
-10.1766  -2.3388   0.1532   2.4074   8.5044

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 25.51232    0.08369 304.856  < 2e-16 ***
cosine      -9.31032    0.11827 -78.721  < 2e-16 ***
sine        -0.69003    0.11843  -5.826 6.96e-09 ***
```

(Figure 3.39)

After fitting an ARIMA (again, Auto ARIMA selects ARIMA(2,1,2)) to the residual, we plot the prediction with the true test value.



(Figure 3.310)

The test RMSE is 1.64, suggesting that fitting a sinusoidal seasonal term effectively captures the seasonal structure and enhances overall model performance.

4. Final Conclusions

This study demonstrates that forecasting daily temperature requires progressively more sophisticated modeling approaches. Baseline models show that the data has strong temporal dependence. Exponential smoothing methods improve predictive accuracy but fail to adequately capture the strong seasonal structure. ARIMA models further improve performance by modeling short-term dependencies. In particular, the hybrid Holt–Winters + ARIMA approach effectively combines seasonal decomposition with residual modeling, achieving strong predictive accuracy.

ARIMAX models using humidity and wind speed lead to additional gains in accuracy, while pressure does not contribute meaningfully. Attempts to include interaction (mixed) terms do not improve generalization and slightly worsen performance, suggesting overfitting. Overall, the most significant improvement is achieved by explicitly modeling seasonality through a sinusoidal term, resulting in the lowest RMSE. These findings highlight that accurately capturing seasonal patterns is more important than increasing model complexity alone, and that external variables play a supportive but secondary role in improving forecasts.

5. Contribution

5.1 Team Formation: Our team formed as a group of people with a shared interest in climate data analysis.

5.2 Author Ordering: Alphabetical by surname.

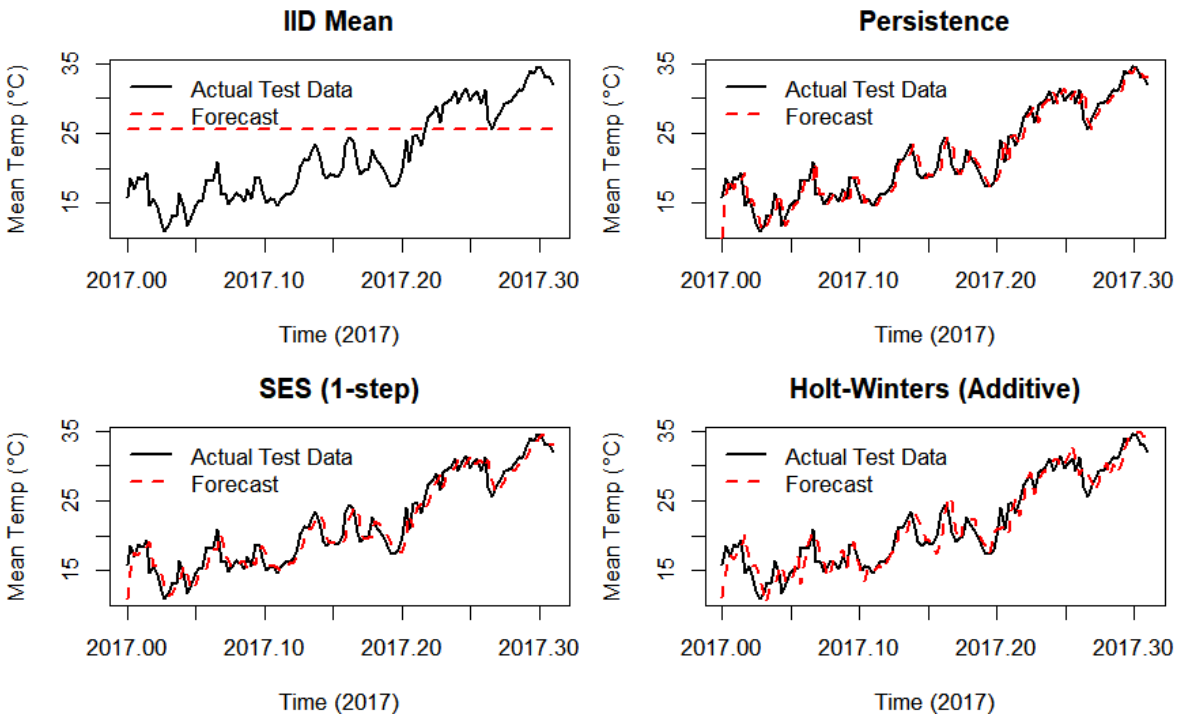
5.3 Contribution Summary

Amy Chen: Contributed to EDA and models for baseline and exponential smoothing methods. This included statistical analysis and implementation in R, as well as assistance with result interpretation and writing the related sections.

Yingying Chu: Led ARIMA modeling, including model identification (ACF/PACF), parameter tuning, coding, and diagnostic analysis. Contributed to methodological discussion and interpretation of results.

Mingyang Gao: Responsible for ARMAX modeling, including incorporating explanatory variables, model specification, and coding. Contributed to comparative analysis and evaluation of model performance.

Shiyu Wu: Coordinated the overall project structure, implemented the hybrid modeling framework, and conducted RMSE-based model comparison. Led integration of results, report writing, and organization of group discussions.

6. Appendix

Complete code and dataset available at https://github.com/mingyang6901-ctrl/443